Academic project carried out at ENP, Algiers

Cascade Speed and Current Control of a Separately-Excited DC Machine

Thyristor-rectifier drive: modelling, analytical controller synthesis and simulation

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Electrical drives and power electronics

A 220 V / 3000 rpm DC drive: six-pulse controlled rectifier, averaged converter model and nested PI/PID loops

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Software

MATLAB / Simulink
Simscape Electrical
Python 3.12 (NumPy, SciPy)

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Abstract

This report presents the complete design and simulation of a cascade speed and current controller for a separately-excited DC machine supplied by a three-phase, six-pulse thyristor rectifier. Each element of the drive is developed in detail: the operation of the controlled bridge and the synthesis of its firing angle, the average-value converter model used for control design, the armature-voltage to speed transfer function and its block-diagram realisation, and the analytical tuning of the current and speed controllers by pole–zero cancellation combined with a settling-time criterion.

A key element of the drive is the firing-angle generator, which applies an inverse-cosine law $\alpha = \arccos(2U_{cm}/U_h - 1)$ to the controller output. In the ideal continuous-conduction model, this law makes the mean bridge voltage an affine function of the control signal. For tuning, the averaged model uses a nominal gain of 22 V per control unit and a first-order lag of 1.667 ms, half the 3.33 ms interval between successive firings. On this linear plant the current PID and the speed PI are tuned in closed form, giving response times of 0.1 s and 1 s respectively, a deliberate ratio of ten that justifies designing the two loops independently.

The project implements the same cascade in two forms. The first uses an averaged converter and motor equations assembled from standard Simulink blocks. The second uses Simscape Electrical to simulate the three-phase supply, individual thyristor firings and the DC machine. An intermediate model with a controlled voltage source lets the physical machine be checked before the bridge is introduced. A separate Python study reproduces the analytical tuning and shows how design choices affect the closed-loop responses.

Keywords: DC machine, cascade control, current limiting, six-pulse thyristor rectifier, firing angle, average-value model, pole–zero cancellation, PID tuning, Simscape Electrical.

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Nomenclature

u_a, i_a	Armature voltage and current	V, A
R_a, L_a	Armature resistance and inductance	Ω, H
L_f	Series smoothing inductance	H
$k\Phi$	Torque and back-EMF constant	V·s/rad
ω	Rotor angular speed	rad/s
J	Rotor inertia	kg·m^2
f_r	Viscous friction coefficient	N·m·s
K_r	Viscous load coefficient	N·m·s
C_r	Load torque	N·m
α	Thyristor firing angle	$^\circ$ or rad
U_{LL}	Line-to-line supply voltage (RMS)	V
V_d, V_{d0}	Mean rectified voltage, and its value at $\alpha = 0$	V
U_{cm}	Current-controller output (control signal)	–
U_h	Firing-law reference amplitude	–
K_c, T_c	Converter static gain and transport lag	V/unit, s
b	Current-loop plant gain, K_c/R_a	A/unit
τ_e, τ_m	Electrical and mechanical time constants	s
$t_{r,i}$, $t_{r,w}$	Target response times, current and speed loops	s
K_p, K_i, K_d	Controller gains	–
ζ, ω_n	Damping ratio and natural frequency	–, rad/s

1 Introduction and system overview

1.1 Why a DC drive is still worth studying

Variable-speed DC drives have largely been displaced by AC drives in new installations, but they remain widespread in existing industrial plant and they retain a strong pedagogical value. The reason is structural: in a separately excited DC machine with constant field, the flux and the armature current are carried by two physically distinct circuits, so the torque is proportional to the armature current with no coupling to compensate and no coordinate transformation to perform. The machine is linear, and every step of the control design can be carried out analytically and checked by hand. The concepts that emerge — nested loops, time-scale separation, current limiting, pole–zero cancellation, the average-value model of a converter — transfer directly to AC drives, where they are harder to see because they are entangled with the vector-control machinery.

This study takes a complete drive, from the mains to the shaft, and develops each of its parts: the controlled rectifier and its firing-angle generator, the averaged converter model, the plant transfer function, and the two nested controllers. The analytical design uses a nominal point of 220 V and 3000 rpm.

1.2 The separately-excited DC machine

The plant is a separately excited DC machine whose speed is regulated through its armature voltage u_a, the field excitation being held constant so that the flux term $k\Phi$ is a constant. Under this condition the machine is linear and is described by an electrical balance on the armature circuit and a mechanical balance on the rotor:

$$u_a(t) = R_a\, i_a(t) + (L_a + L_f)\frac{\mathrm{d}i_a}{\mathrm{d}t} + k\Phi\,\omega(t), \tag{1}$$

$$J\frac{\mathrm{d}\omega}{\mathrm{d}t} = k\Phi\, i_a(t) - (f_r + K_r)\,\omega(t), \tag{2}$$

where $k\Phi\,\omega$ is the back-EMF, $k\Phi\, i_a$ the electromagnetic torque, f_r the mechanical friction and K_r a viscous load coefficient.

The inductance L_f deserves a comment because it dominates the electrical dynamics. It is not a machine parameter but a smoothing inductor deliberately placed in series with the armature to reduce the current ripple produced by the rectifier. Its value of 276 mH is more than twenty-five times the armature inductance itself, so the electrical time constant of the drive is essentially set by a component that the designer chooses rather than by the machine. This is a recurring feature of thyristor drives: the filter is part of the plant.

Eliminating the current between the two balances gives the armature-voltage to speed transfer function that underpins the whole study:

$$G(s) = \frac{\Omega(s)}{U_a(s)} = \frac{k\Phi}{(R_a + (L_a{+}L_f)s)\,(Js + f_r + K_r) + k\Phi^2}. \tag{3}$$

Its derivation and block-diagram realisation are given in Section 3.

1.3 Cascade control architecture

The controller uses the classical cascade structure with two nested loops. An inner current loop regulates the armature current i_a and is designed to be fast; its reference is produced

Table 2. Parameters of the studied DC machine, confirmed against the Simscape *DC Machine* block.

Symbol	Quantity	Value	Unit
R_a	armature resistance	2.33	Ω
L_a	armature inductance	10.6	mH
L_f	series smoothing inductance	276	mH
$k\Phi$	torque / back-EMF constant	0.57	V·s/rad
J	rotor inertia	7.5×10^{-3}	kg·m^2
f_r	viscous friction	1.75×10^{-4}	N·m·s
k	equivalent speed-loop scaling (design)	1.25×10^{-2}	–

by an outer speed loop, designed to be comparatively slow. A saturation block placed on the current reference provides natural current limiting during large transients such as start-up and load steps, which protects both the machine and the thyristors.

The design deliberately enforces a time-scale separation: the current loop is tuned to respond about ten times faster than the speed loop, with target response times of 0.1 s and 1 s respectively. Because the inner loop settles long before the outer loop reacts, the speed controller effectively sees the current loop as an ideal, instantaneous torque actuator, and the two loops can be designed independently. This is the assumption on which Section 4 rests, and it is the reason the ratio of ten is a design constraint rather than a convenience.

The complete signal chain is therefore

$$\omega_{\text{ref}} \rightarrow \boxed{\text{Speed PI}} \rightarrow i_{a,\text{ref}} \rightarrow \boxed{\text{Current PID}} \rightarrow U_{cm} \rightarrow \boxed{\text{Rectifier}} \rightarrow u_a \rightarrow \boxed{\text{DC machine}} \rightarrow \omega, \tag{4}$$

with i_a fed back to the inner summing junction and ω to the outer one.

2 The controlled rectifier and its model

The armature is supplied by a three-phase, six-pulse thyristor rectifier, that is, a fully controlled Graetz bridge. This section explains how the bridge works, how its firing angle is generated, and how the whole converter is condensed into the single first-order transfer function used for control design.

2.1 The six-pulse bridge

The bridge is a Simscape *Universal Bridge* configured with three arms and six thyristors, fed by a three-phase source of line-to-line voltage $U_{LL} = 220\,\text{V}$ RMS at 50 Hz. The six thyristors conduct in pairs, each for 120°, and there are six commutations per mains period. The rectified voltage therefore carries a ripple at

$$f_{\text{ripple}} = 6 \times 50 = 300\ \text{Hz}, \tag{5}$$

which the 276 mH smoothing inductor filters into a nearly constant armature current.

Unlike a diode, a thyristor conducts only after its gate has received a pulse, so conduction can be delayed by a controllable firing angle α measured from the natural commutation instant. That delay is the control handle of the entire drive.

2.2 Mean output voltage and the nonlinearity it introduces

Averaging the six-pulse waveform over one 60° segment gives the classical result for the mean output of a fully controlled bridge:

$$V_d(\alpha) = \frac{3\sqrt{2}}{\pi} U_{LL} \cos\alpha = \frac{3\sqrt{6}}{\pi} V_\varphi \cos\alpha \approx 1.35\, U_{LL} \cos\alpha, \tag{6}$$

where $V_\varphi = U_{LL}/\sqrt{3}$ is the phase RMS voltage. With $U_{LL} = 220\,\text{V}$ the no-delay maximum is

$$V_{d0} = 1.35 \times 220 \approx 297\ \text{V}, \qquad V_d(\alpha) = 297 \cos\alpha. \tag{7}$$

Equation (6) gives the two possible average voltage polarities under continuous conduction. $\alpha < 90°$ gives $V_d > 0$ and the bridge rectifies, $\alpha > 90°$ gives $V_d < 0$ and the bridge inverts, which allows regenerative braking, and $\alpha = 90°$ gives $V_d = 0$. It also shows the difficulty: V_d is a cosine of the control variable, so the converter gain $\mathrm{d}V_d/\mathrm{d}\alpha = -V_{d0} \sin\alpha$ varies from zero at $\alpha = 0$ to its maximum at $\alpha = 90°$. A controller tuned for one operating point would be mistuned at another. This is what the firing-angle generator exists to fix.

2.3 The firing-angle generator: a linearising control law

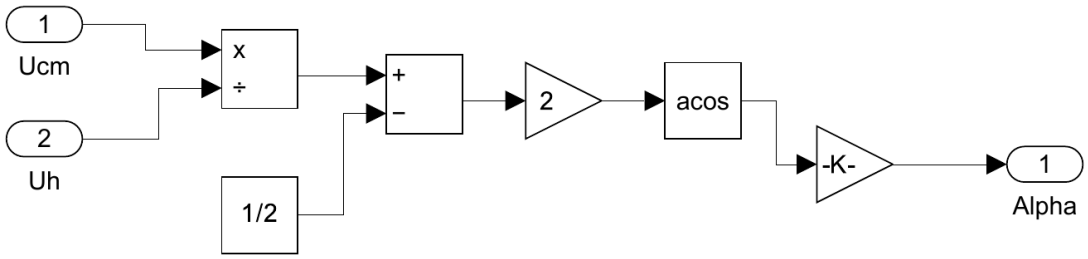


Figure 1. The firing-angle generator subsystem. It computes $\alpha = \arccos(2U_{cm}/U_h - 1)$, the inverse-cosine law that linearises the ideal mean-voltage characteristic of the thyristor bridge.

The generator of Figure 1 converts the current controller's output U_{cm} into the firing angle in degrees supplied to the six-pulse pulse generator. Reading it from left to right, with $U_h = 10$ as the reference amplitude:

$$\frac{U_{cm}}{U_h} \xrightarrow{\times 2,\,-1} \left(\frac{2U_{cm}}{U_h} - 1\right) \xrightarrow{\arccos} \alpha_{\text{rad}} \xrightarrow{\times 180/\pi} \alpha_{\text{deg}}, \tag{8}$$

that is,

$$\boxed{\alpha = \arccos\left(\frac{2U_{cm}}{U_h} - 1\right).} \tag{9}$$

The reason for the inverse cosine is that it cancels the cosine of the bridge exactly. Substituting (9) into (6),

$$V_d = V_{d0} \cos\alpha = V_{d0} \left(\frac{2U_{cm}}{U_h} - 1\right), \tag{10}$$

so the mean output voltage becomes an affine function of the control signal. The mapping is $U_{cm} = 0 \Rightarrow \alpha = 180° \Rightarrow V_d = -V_{d0}$, $U_{cm} = 5 \Rightarrow \alpha = 90° \Rightarrow V_d = 0$, and $U_{cm} = 10 \Rightarrow \alpha = 0° \Rightarrow V_d = +V_{d0}$. Within this command interval the ideal converter has a constant slope of

$2V_{d0}/U_h$. A six-pulse generator with a 60° pulse width places the gate pulses at the requested angle using a reference synchronised with the simulated supply.

2.4 The average-value model, and why a transfer function

For control design the switching bridge cannot be used directly. It is nonlinear before linearisation and discrete in time, since it can only act at the six firing instants per period, whereas pole–zero placement and settling-time design require a linear, continuous plant. The standard remedy is the average-value model, which replaces the whole converter — bridge, firing law and pulse generator — by two effects.

A static gain. The block-diagram model uses a separate nominal scaling for controller design. A command of ten units corresponds to the rated armature voltage. This scaling and its zero offset are specific to the averaged model:

$$K_c = \frac{U_{a,\text{rated}}}{U_{cm,\max}} = \frac{220}{10} = 22 \qquad [\text{V per control unit}]. \tag{11}$$

A small transport lag. After the controller changes U_{cm}, the new firing angle can only take effect at the next thyristor firing. Firings occur every $1/(6{\times}50) = 3.33\,\text{ms}$, and a change arriving at a uniformly distributed instant within that interval waits on average half of it:

$$T_c = \frac{1}{2 \times 6 \times 50} = 1.667\ \text{ms}, \tag{12}$$

modelled as a first-order lag $1/(T_c s + 1)$.

Combining the two, the rectifier is represented by the single first-order block that appears in the Simulink block diagram:

$$\frac{U_a(s)}{U_{cm}(s)} = \frac{K_c}{T_c s + 1} = \frac{22}{T_c s + 1}. \tag{13}$$

The advantage of a transfer function is that it turns the converter into a linear element which can be folded into the loop transfer function, so that the current PID can be tuned in closed form (Section 4). The switching model is kept separately, in Section 6, in order to verify that the controllers designed on the averaged model still behave correctly once the real ripple, the dead time and the discrete firing are present. This is the same methodological choice as in any converter study: design on the average, verify on the switching model.

The static gain also fixes the plant gain of the current loop:

$$b = \frac{K_c}{R_a} = \frac{22}{2.33} = 9.442 \qquad [\text{A per control unit}], \tag{14}$$

that is, control to voltage through ×22, then voltage to current through $\times 1/R_a$.

3 Plant transfer function and its block-diagram realisation

3.1 Derivation

Taking Laplace transforms of the two balances of Section 1, with the load written as a viscous term $C_r = K_r\,\omega$:

$$U_a = (R_a + (L_a{+}L_f)s)I_a + k\Phi\,\Omega, \qquad Js\,\Omega = k\Phi\,I_a - (f_r + K_r)\Omega. \tag{15}$$

Eliminating I_a returns (3). The denominator is second order, as expected for a system with one electrical and one mechanical energy store, and the term $k\Phi^2$ is the contribution of the back-EMF, which acts as an internal negative feedback path and is what makes an unregulated DC machine self-stabilising.

3.2 The block-diagram model of the machine

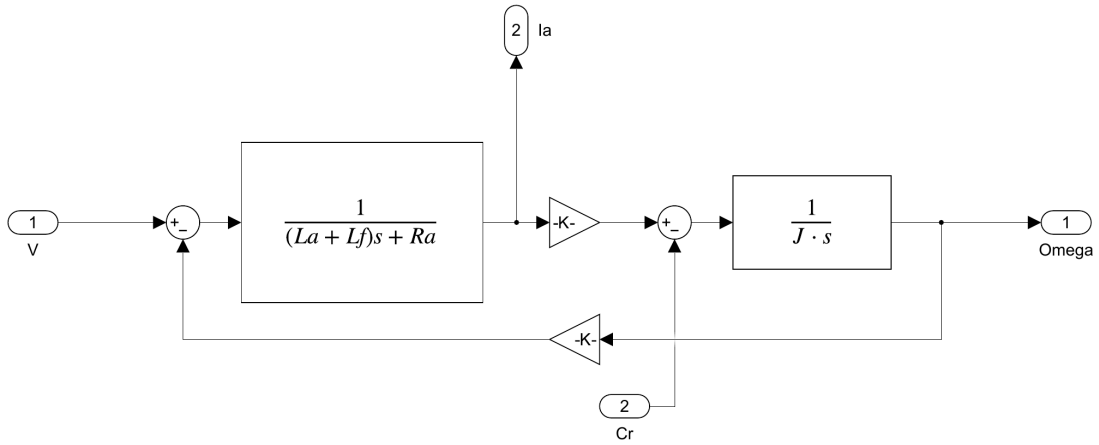


Figure 2. The *DC machine* subsystem: the armature admittance $1/((L_a{+}L_f)s + R_a)$ produces the current, $k\Phi$ converts it to torque, $1/(Js)$ integrates to speed, and $k\Phi\,\omega$ is fed back as the back-EMF.

The subsystem of Figure 2 realises (3) as an explicit block diagram rather than as a transfer-function block. This is not a cosmetic choice: it makes the internal current signal available for measurement, which is required by the inner loop, and it exposes the back-EMF path so that its role can be seen. Following the diagram, the armature voltage V has the back-EMF $k\Phi\,\omega$ subtracted, drives the armature admittance to give the current I_a, which is scaled by $k\Phi$ into torque; the load torque C_r is subtracted and the net torque is integrated by $1/(Js)$ into the speed:

$$I_a = \frac{V - k\Phi\,\Omega}{(L_a + L_f)s + R_a}, \qquad \Omega = \frac{k\Phi\,I_a - C_r}{Js}. \tag{16}$$

Substituting one into the other returns $G(s)$ exactly, which confirms that the block diagram and the analytical model agree.

3.3 Fixing the operating point

The load is not a free parameter. Since the study fixes both the armature voltage and the resulting speed, the viscous load coefficient K_r must be back-computed so that the requested operating point is exactly the machine's steady state. Setting $\mathrm{d}i_a/\mathrm{d}t = \mathrm{d}\omega/\mathrm{d}t = 0$ in (1) and (2) and solving for K_r:

$$K_r = \frac{1}{R_a}\left(\frac{k\Phi\,U_a}{\omega^\star} - k\Phi^2\right) - f_r. \tag{17}$$

With $U_a = 220\,\mathrm{V}$ and $\omega^\star = 3000 \times 2\pi/60 = 314.16\,\mathrm{rad/s}$ this gives $K_r = 0.03170$, a DC gain $G(0) = 1.428\,00\,\mathrm{rad{\cdot}s^{-1}{\cdot}V^{-1}}$ and hence $\omega_{ss} = G(0) \times 220 = 3000.0\,\mathrm{rpm}$. The construction is self-consistent, which is a useful check that no unit conversion has been dropped.

Two characteristic time constants follow and are used in the tuning:

$$\tau_e = \frac{L_a + L_f}{R_a} = 0.1230\ \mathrm{s}, \qquad \tau_m = \frac{J R_a}{k\Phi^2} = 0.0538\ \mathrm{s}. \tag{18}$$

It is worth pausing on their relative magnitude. In most small DC machines the electrical time constant is far shorter than the mechanical one, and the standard cascade argument, which assumes the current settles before the speed moves, follows immediately. Here τ_e is more than twice τ_m, because the 276 mH smoothing inductor has made the electrical circuit unusually slow. The time-scale separation therefore does not come from the machine: it has to be created by the controllers, and this is precisely what the choice $t_{r,i} = 0.1\,\mathrm{s}$ against $t_{r,w} = 1\,\mathrm{s}$ does. The current controller must actively speed the electrical dynamics up.

4 Controller design

Both controllers are implemented with the Simulink *PID* block in parallel form with the two-degree-of-freedom option disabled, $C(s) = K_p + K_i/s + K_d\,\dfrac{N}{1 + N/s}$ with a derivative filter coefficient $N = 100$; the speed loop uses only the PI terms.

They are tuned analytically using two ideas. The first is pole–zero cancellation: the controller zeros are placed on the plant poles so that the open loop becomes a pure integrator, whose closed loop is a first-order lag of exactly known time constant. The second is a settling-time criterion: a first-order lag of time constant τ reaches 99 % of its final value after $t_s = -\ln(0.01)\,\tau \approx 4.6\,\tau$, which is the source of the recurring factor 4.6.

4.1 Inner current loop

Neglecting the back-EMF coupling, which is legitimate because the speed varies far more slowly than the current within the bandwidth of this loop, the plant from control signal to current is the converter lag in series with the armature electrical lag:

$$P_i(s) = \frac{b}{(1 + \tau_e s)(1 + T_c s)}, \qquad b = \frac{22}{R_a} = 9.442. \tag{19}$$

Writing the PID with its zeros exposed and choosing

$$\frac{K_p}{K_i} = \tau_e + T_c, \qquad \frac{K_d}{K_i} = \tau_e\,T_c \tag{20}$$

makes its numerator factor as $(1 + \tau_e s)(1 + T_c s)$, which cancels both plant poles. The open loop collapses to $L(s) = K_i\,b/s$ and the closed loop is first order with $\tau_{cl} = 1/(K_i b)$. Imposing $t_s = 4.6\,\tau_{cl} = t_{r,i}$ fixes the gains:

$$K_i = \frac{4.6}{t_{r,i}\,b}, \qquad K_p = (\tau_e + T_c)\,K_i, \qquad K_d = \tau_e\,T_c\,K_i. \tag{21}$$

For $t_{r,i} = 0.1\,\mathrm{s}$ this gives $K_i = 4.872$, $K_p = 0.607$ and $K_d = 0.001$. The derivative term is

very small because T_c is very small, so in practice the controller is a PI with a minor correction for the converter dead time. It is worth noting that this is exactly what one would expect: the derivative action exists only to cancel the fastest pole, and that pole is nearly at infinity on the time scale of the loop.

4.2 Outer speed loop

With the current loop fast and close to unity gain within its bandwidth, the mechanical plant seen by the speed controller reduces to a first-order lag between current demand and speed,

$$P_\omega(s) = \frac{a}{1 + \frac{J}{k}s}, \qquad a = \frac{k\Phi}{k} = 45.6, \tag{22}$$

where $k = 1.25 \times 10^{-2}$ is the equivalent damping and measurement scaling used in the models. A PI controller cancels the mechanical pole when $K_p/K_i = J/k$, again reducing the open loop to $K_i a/s$; imposing $t_s = 4.6/(K_i a) = t_{r,w}$ gives

$$K_i = \frac{4.6}{t_{r,w}\, a}, \qquad K_p = \frac{J}{k}\, K_i. \tag{23}$$

For $t_{r,w} = 1\,\mathrm{s}$ this gives $K_i = 0.1009$ and $K_p = 0.0605$.

Table 3. Analytically tuned controller gains.

Loop	Type	K_p	K_i	K_d	Target response
Current (inner)	PID	0.607	4.872	0.001	$t_{r,i} = 0.1\,\mathrm{s}$
Speed (outer)	PI	0.0605	0.1009	—	$t_{r,w} = 1\,\mathrm{s}$

4.3 Units, scaling and limiting

In the models the speed is measured through a scaling gain $k = 1.25 \times 10^{-2}$, and the reference and display use the conversions $60/2\pi$ from rad/s to rpm and $2\pi/60$ in the opposite direction. These factors have no dynamic effect but they must be applied consistently, since a missing conversion changes the effective loop gain by a factor of about 9.55 and would silently invalidate the tuning.

The current reference is hard-limited by a saturation block. This is the element that makes the cascade worth building in the first place: without it, a large speed step would demand an arbitrarily large current, which the thyristors would not survive. With it, a large step simply produces a constant-current, and therefore constant-torque, acceleration until the speed error becomes small enough for the loop to return to its linear range.

4.4 An alternative tuning for constant load torque

When the load torque is constant rather than viscous, the speed loop can be tuned instead by direct second-order pole placement with a chosen damping ratio ζ and natural frequency ω_n:

$$K_p = \frac{2\zeta\omega_n J}{k\Phi}, \qquad K_i = \frac{\omega_n^2 J}{k\Phi}, \tag{24}$$

the current loop then using $\tau_m = 53.8\,\text{ms}$ in place of $\tau_e + T_c$. This variant is analysed in Section 5 because it makes the separate roles of ζ and ω_n directly visible, which the pole–zero cancellation design deliberately hides by forcing a first-order response.

5 Python verification and parametric study

The MATLAB design was re-implemented independently in Python, using NumPy, SciPy and Matplotlib, for two reasons: to confirm the tuning on a platform that shares no code with the Simulink models, and to visualise how the closed-loop response depends on the design targets t_r, ζ and ω_n.

5.1 Cross-check of the design quantities

Table 4. Design quantities: MATLAB versus the independent Python implementation.

Quantity	MATLAB	Python
Load coefficient K_r	0.031697	0.031697
DC gain $G(0)$	1.428 00	1.428 00
Steady-state speed	3000.0 rpm	3000.0 rpm
Current PID $K_p/K_i/K_d$	0.607 / 4.872 / 0.001	0.607 / 4.872 / 0.001
Speed PI K_p/K_i	0.0605 / 0.1009	0.0605 / 0.1009

The two implementations agree to the printed precision on every quantity. This is the minimum that should be expected of a correct port, but it is worth recording, because a disagreement here would have pointed to a unit-conversion error or to a different interpretation of K_r, both of which are easy to make.

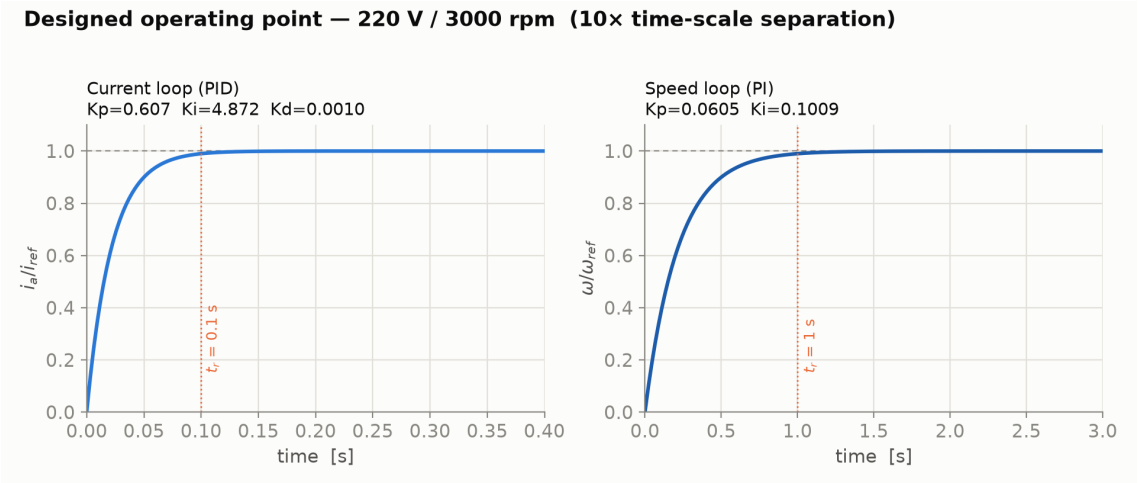


Figure 3. Step responses at the designed operating point: inner current loop (left) and outer speed loop (right), showing the deliberate ten-to-one time-scale separation.

Figure 3 shows the two loops at the designed operating point. The current loop settles in about 0.1 s and the speed loop in about 1 s, exactly as specified, and neither exhibits overshoot: pole–zero cancellation makes both closed loops first order by construction.

5.2 First order: the response time sets the rise time

After pole–zero cancellation the current loop is exactly first order, $T(s) = 1/(1 + \tau s)$ with $\tau = t_{r,i}/4.6$. Sweeping the target $t_{r,i}$ slides the single closed-loop pole along the real axis and therefore scales the rise time, with no overshoot at any setting.

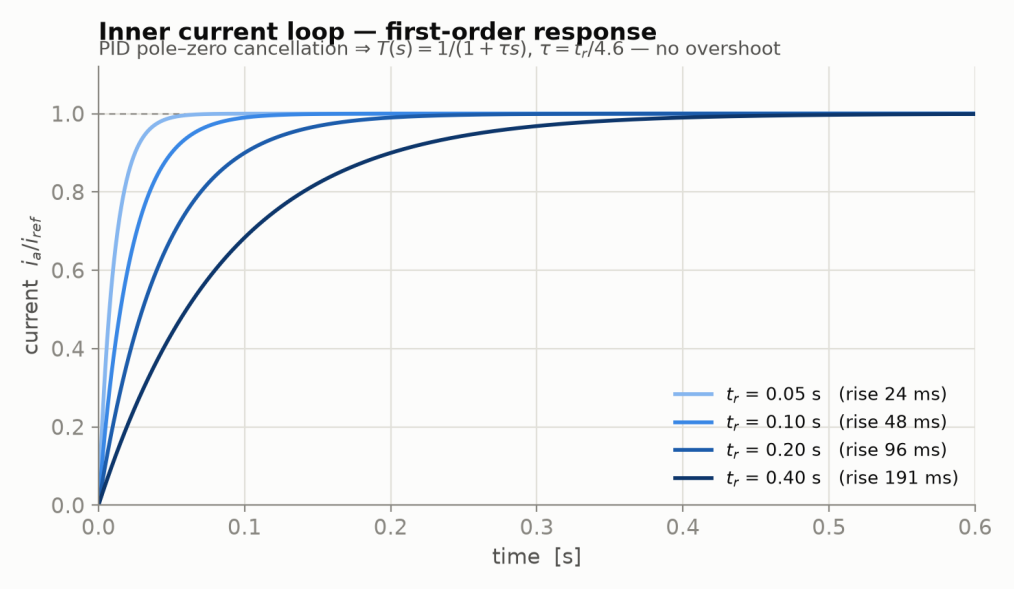


Figure 4. First-order current loop: the response-time target sets the rise time, with no overshoot at any setting.

Table 5. First-order current loop: rise time versus the response-time target.

$t_{r,i}$	Closed-loop τ	10–90 % rise time
0.05 s	10.9 ms	24 ms
0.10 s	21.7 ms	48 ms
0.20 s	43.5 ms	96 ms
0.40 s	87.0 ms	191 ms

The measured rise times of Table 5 are consistently 2.2 times the closed-loop time constant, which is the theoretical value $\ln 9 \approx 2.197$ for a first-order system. The agreement confirms that the cancellation is exact in the model and that no unmodelled pole is interfering.

5.3 Second order: the damping ratio sets the overshoot

For the constant-torque design of Section 4.4 the speed loop is second order,

$$T(s) = \frac{2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \tag{25}$$

Holding $\omega_n = 5\,\mathrm{rad/s}$ and sweeping ζ moves the closed-loop poles from lightly damped to overdamped.

The overshoot is higher than the familiar textbook values, 21 % instead of about 4 % at $\zeta = 0.707$, and even the critically damped case overshoots by 14 % instead of not at all. The explanation is the numerator: a PI controller contributes a finite zero at $s = -\omega_n/(2\zeta)$, and the

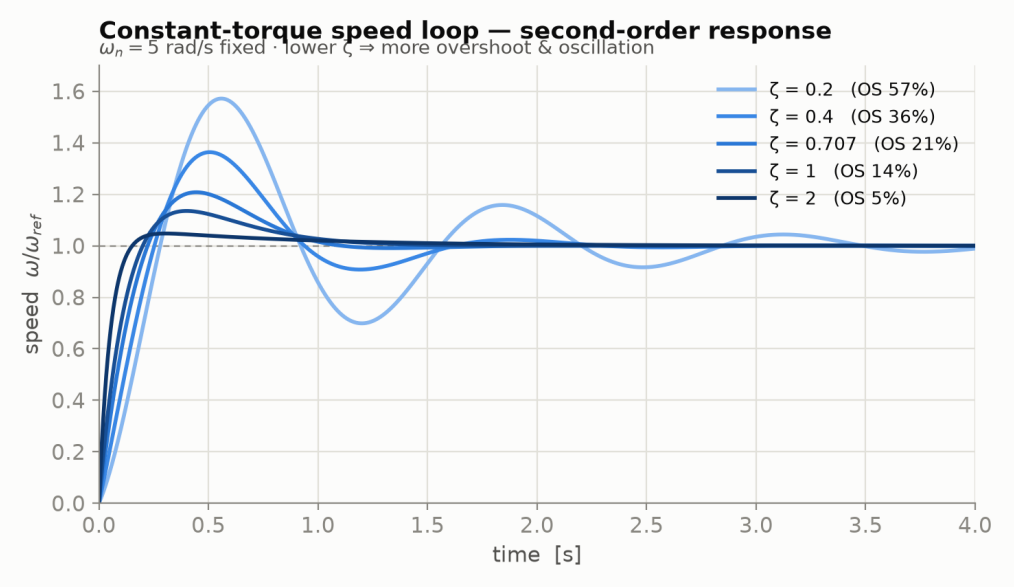


Figure 5. Second-order speed loop: the damping ratio sets the overshoot and the oscillation, at fixed $\omega_n = 5\,\mathrm{rad/s}$.

Table 6. Second-order speed loop: overshoot versus damping ratio.

ζ	Closed-loop poles	Overshoot
0.2	complex, lightly damped	57 %
0.4	complex	36 %
0.707	complex	21 %
1.0	critically damped, real, equal	14 %
2.0	overdamped, real, distinct	5 %

closed-loop transfer function above is therefore not the canonical second-order form found in textbooks, which has a constant numerator. The zero differentiates part of the reference, which speeds the rise but amplifies the first peak. This is precisely why the nominal design of Section 4 prefers pole–zero cancellation, which produces a first-order response with no overshoot at all.

5.4 Second order: the natural frequency sets the settling time

Holding $\zeta = 0.7$ and sweeping ω_n preserves the shape of the response while scaling its time axis, so the settling time is inversely proportional to ω_n.

Table 7. Second-order speed loop: settling time versus natural frequency.

ω_n	$\pm 2\,\%$ settling time
2 rad/s	2.44 s
5 rad/s	0.98 s
10 rad/s	0.49 s
20 rad/s	0.24 s

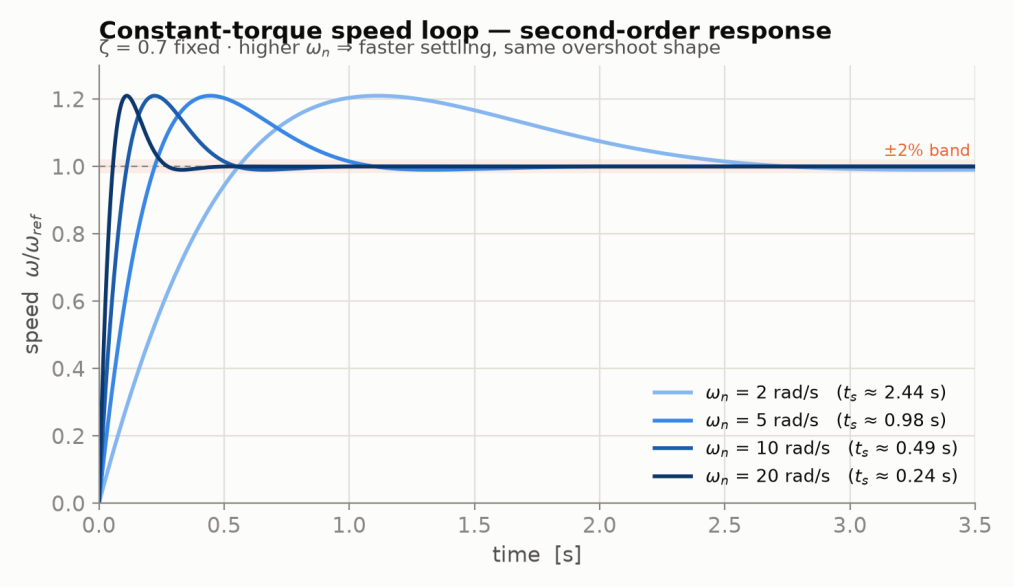


Figure 6. Second-order speed loop: the natural frequency sets the settling time, at fixed $\zeta = 0.7$.

The product $\omega_n t_s$ is constant at about 4.9 across the whole sweep, as the theory requires. The two parameters therefore act as independent design handles: ω_n is the bandwidth, or speed, handle and ζ is the damping, or overshoot, handle. In the real machine ω_n is ultimately capped by the bandwidth of the inner current loop, which is the reason the cascade needs its time-scale separation and the reason a bandwidth cannot simply be increased at will.

6 Two implementations in MATLAB/Simulink

6.1 Implementation 1. Averaged Simulink block diagram

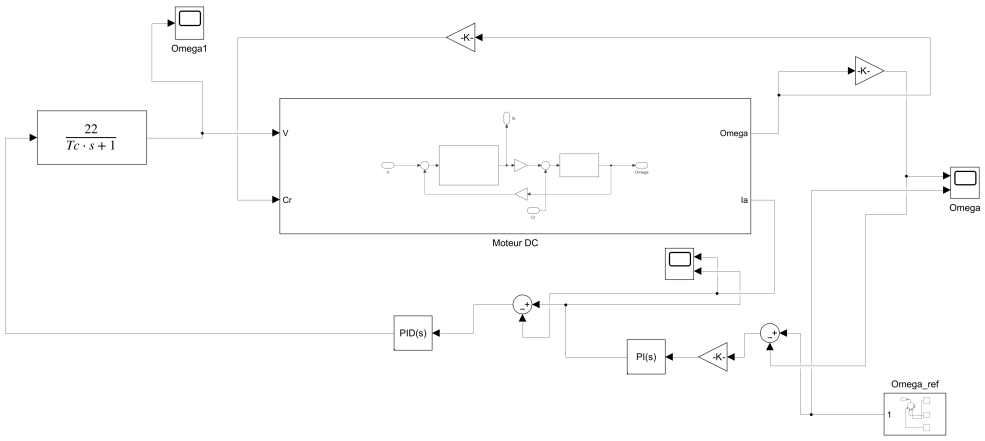


Figure 7. Simulink cascade with the *Moteur DC* subsystem and averaged converter.

The speed PI turns speed error into a current request. The current PID compares it with the modelled armature current and sends its output through $22/(T_c s + 1)$ to approximate the converter voltage and firing delay. Inside the motor subsystem, back-EMF affects current, current produces torque, and the integral of net torque divided by inertia gives speed.

The model requests 1,000, 3,000 and 1,500 rpm at 1, 2 and 3 s. It peaks at 3,046.88 rpm and 13.08 A (Figure 8). The Simscape test uses different step timing and includes a current-reference limiter.

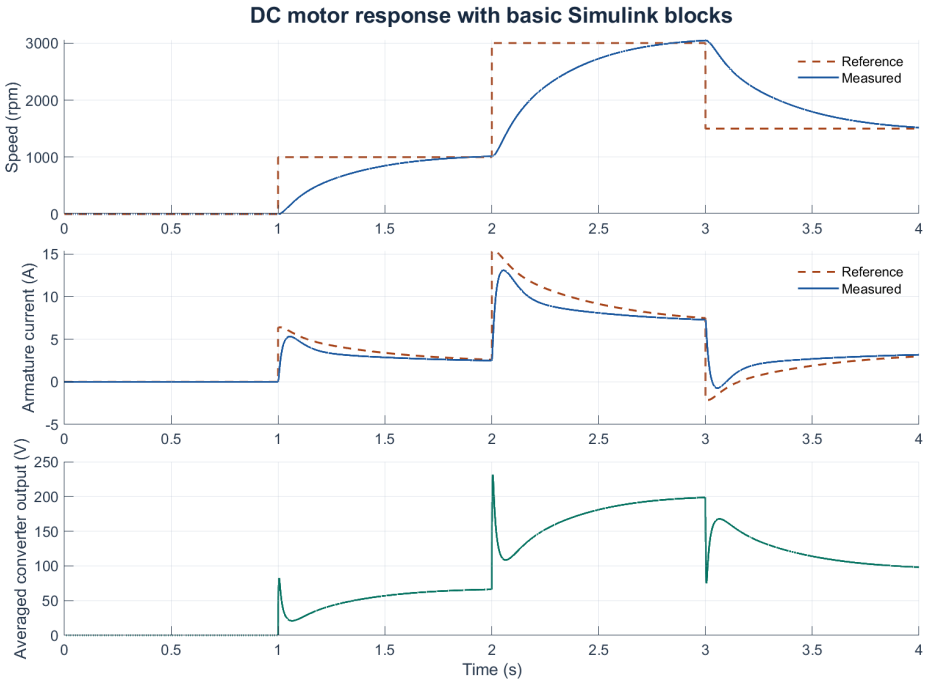


Figure 8. Speed, armature current and average converter voltage during the 4 s test.

6.2 Implementation 2. Switched Simscape Electrical drive

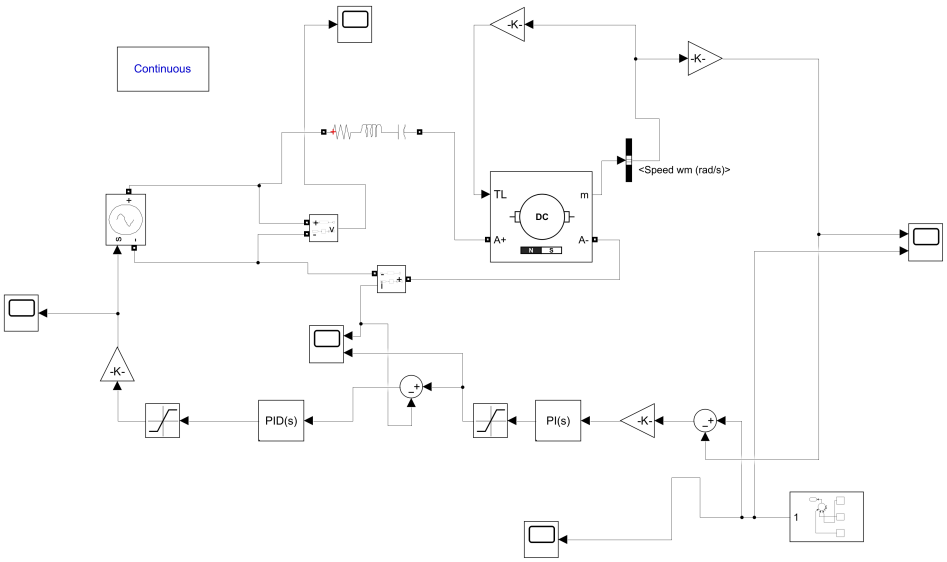


Figure 9. Intermediate check with a library DC Machine and a controlled voltage source. This setup separates the change of machine model from the introduction of thyristor switching.

I first replaced the equation blocks with the *DC Machine* block from Specialized Power Systems and connected the external 276 mH smoothing inductor. A controlled voltage source supplied the armature circuit while I checked the speed feedback, the current sensor and the load-torque input. The machine block uses its permanent-magnet setting with the same fixed torque constant as the analytical model, which represents the constant-field operating assumption.

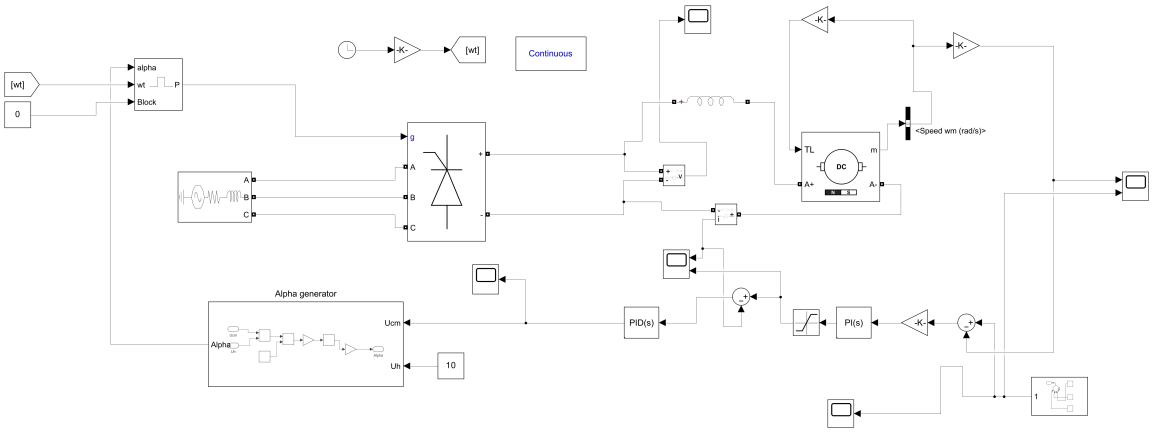


Figure 10. Switched Simscape Electrical model. The six-pulse generator drives the thyristors in the Universal Bridge. The bridge feeds the DC machine through the series smoothing inductor.

I then replaced the controlled source with a 220 V line-to-line RMS, 50 Hz three-phase supply and a *Universal Bridge* configured for six thyristors. The speed PI and current PID retain their roles and gains. The switched model limits the current request to ± 7 A, and the inner PID output now goes to the inverse-cosine angle generator. A six-pulse block converts this angle into gate pulses. The measured armature current and shaft speed feed back to the controllers. The voltage probe sits across the DC output of the bridge, upstream of the smoothing inductor.

6.3 Simscape simulation results

The switched model was run for 20 s in MATLAB R2024b with *powergui* in continuous mode. The speed reference changes to 1,000, 3,000 and 1,500 rpm at 1, 5 and 10 s, respectively, under a viscous load proportional to shaft speed. Figure 11 plots the measured speed and armature current against their references, together with the electromagnetic torque from the machine block. The speed peaks at 3,004.42 rpm and averages 1,500 rpm in the last 0.2 s. The current reference is limited to 7 A during acceleration, and the measured current briefly peaks at 7.07 A.

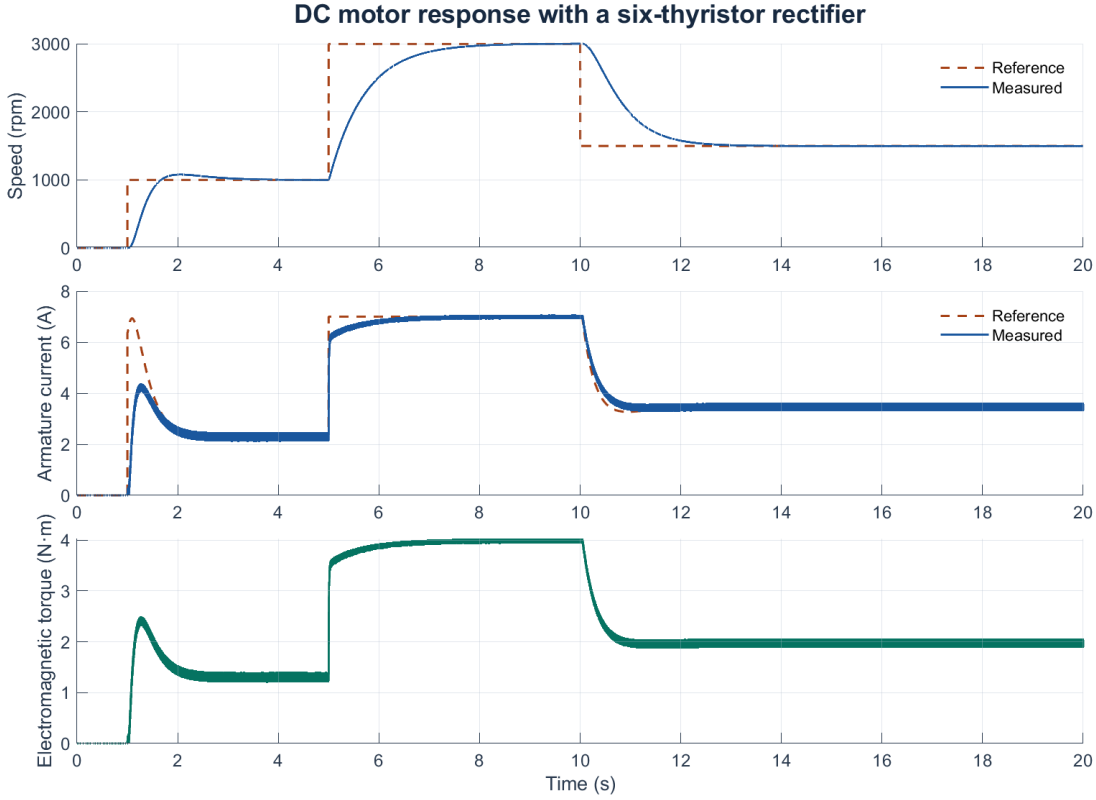


Figure 11. Switched Simscape model. Speed, armature current and electromagnetic torque from the simulation with three reference steps.

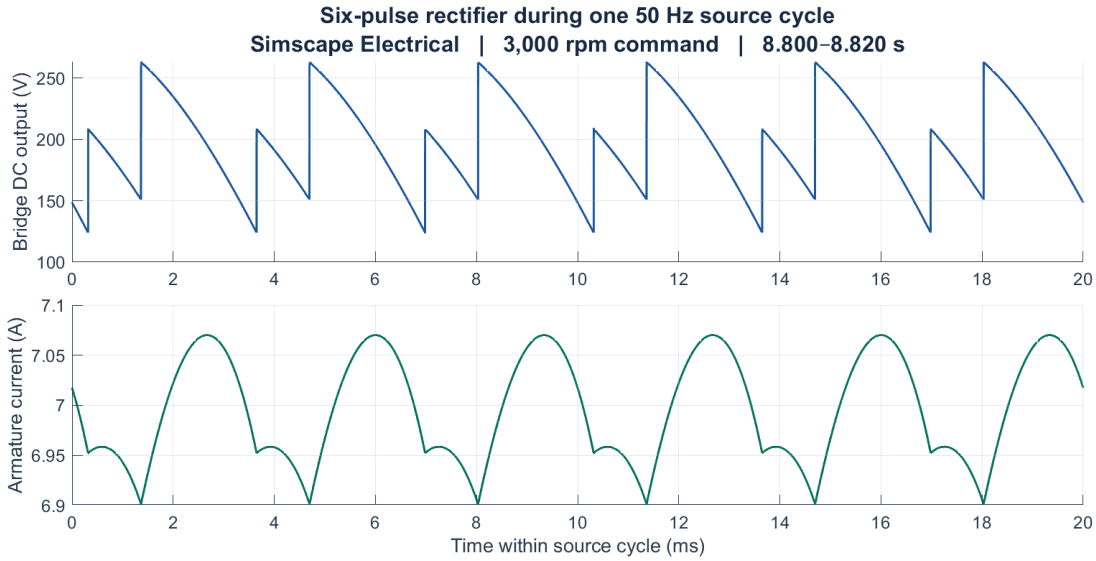


Figure 12. Simscape detail at the 3,000 rpm operating plateau. The voltage probe shows the six-pulse bridge output upstream of the smoothing inductor. The armature current is measured in series with the machine.

The close-up in Figure 12 shows why it is useful to simulate the switches. A 50 Hz supply gives six voltage pulses every 20 ms, or a pulse frequency of 300 Hz. From 8.800 to 8.820 s the bridge voltage ranges from 123.90 to 263.20 V, while the armature current stays between 6.90 and 7.07 A. The series inductance opposes rapid changes in current.

7 Discussion and limitations

Current direction. A single fully controlled bridge can deliver positive or negative average DC voltage with continuous current of one polarity. With a fixed field, reversing the electromagnetic torque at the same speed requires reversing the armature current. A dual converter with anti-parallel bridges can support both current directions.

Constant field. The whole analysis relies on $k\Phi$ being constant, which restricts operation to the base-speed region. Operation above base speed would require field weakening, adding a second, slow control loop on the field current and making $k\Phi$ a variable in every equation of Section 3.

The load model. The load has been represented as purely viscous, with K_r back-computed from the desired operating point through (17). This is a convenient choice because it makes the operating point exact by construction, but it is also a strong assumption: a real load with constant torque, or with a torque proportional to the square of the speed as in a fan, would change the small-signal gain of the plant and therefore the effective loop tuning. The alternative design of Section 4.4 exists precisely to cover the constant-torque case.

Neglected commutation overlap. The averaged model of Equation (6) assumes instantaneous commutation. In a real bridge the source inductance causes an overlap angle during which two thyristors in the same group conduct simultaneously, which reduces the mean output voltage by a term proportional to the load current. This appears as an additional apparent

resistance in series with the armature, and at high current it would shift the operating point slightly below the predicted value.

8 Conclusion

A cascade speed and current controller for a thyristor-fed separately excited DC machine has been developed in full detail. The controlled rectifier was analysed from first principles, from the six-pulse bridge and its cosine voltage law to the firing-angle generator whose inverse-cosine law linearises the converter, and was then condensed into the average-value model $22/(T_c s + 1)$, whose gain and lag were both derived from the drive's own quantities rather than assumed.

On that linear plant, the PI and PID gains were obtained analytically by pole–zero cancellation combined with a settling-time criterion. The resulting design has transparent and independently adjustable targets: one number sets the current-loop response time, another sets the speed-loop response time, and the ratio between them is the time-scale separation that makes the cascade valid. The same framework also accommodates the alternative constant-torque design, in which the roles of the damping ratio and of the natural frequency become explicit.

The averaged model shows how the cascade follows from the machine equations. The switched model exposes effects that the first-order converter approximation does not represent, including discrete firings and the voltage pulses at the bridge output. The comparison with the armature current makes the smoothing inductor's role visible. The independent Python results document the analytical gain calculations and the effect of the response-time targets.

A Summary of the design computations

Table 8 collects the whole design chain in the order in which the quantities are computed, so that the numerical results can be reproduced.

Table 8. Design chain, in computation order.

Step	Expression	Value	Unit
Target speed in rad/s	$3000 \times 2\pi/60$	314.16	rad/s
Load coefficient	Eq. (17)	0.031 697	N·m·s
DC gain	$G(0)$	1.428 00	rad·s^{-1}/V
Steady-state check	$G(0) \times 220$	3000.0	rpm
Electrical time constant	$(L_a + L_f)/R_a$	0.1230	s
Mechanical time constant	$JR_a/k\Phi^2$	0.0538	s
Converter gain	220/10	22	V/unit
Converter lag	$1/(2 \times 6 \times 50)$	1.667	ms
Current-loop plant gain	K_c/R_a	9.442	A/unit
Current K_i	$4.6/(t_{r,i}\, b)$	4.872	–
Current K_p	$(\tau_e + T_c)K_i$	0.607	–
Current K_d	$\tau_e T_c K_i$	0.001	–
Speed-loop gain	$k\Phi/k$	45.6	–
Speed K_i	$4.6/(t_{r,w}\, a)$	0.1009	–
Speed K_p	$(J/k)K_i$	0.0605	–

B Organisation of the accompanying models

Table 9. Organisation of the accompanying code and models.

Path	Contents
`MATLAB/Main & Setup/`	`main.m`, `dc_machine_params.m`, `dc_tf_function.m`
`MATLAB/PID Block Diagram/`	transfer-function block-diagram model
`MATLAB/Library/`	Simscape Electrical model with the thyristor bridge
`MATLAB/assets/`	model diagrams and Simscape result figures
`Python/`	independent port reproducing the tuning and the sweeps
`Python/results/`	the response and parametric figures of Section 5

To reproduce the results, open `main.m`, set the machine preset, the target armature voltage and the target speed, and run it. The script prints the tuned gains and populates the workspace, after which any of the Simulink models can be opened and run. The Python scripts are self-contained and regenerate the figures of Section 5 directly.

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Authorship note

I carried out this project at ENP and contributed my own ideas to the report. I used ChatGPT to help rewrite the text.